Simplification and Differentiation of a composite Logarithmic Function.

$$f(x) = \ln\left(\frac{(x+1)^3}{\sqrt[4]{x^2+1}}\right)$$

$$f(x) = \ln(x+1)^3 - \ln\left(\sqrt[4]{x^2+1}\right)$$

$$f(x) = 3\ln(x+1) - \frac{1}{4}\ln(x^2+1)$$

$$\Rightarrow f'(x) = 3\frac{1}{(x+1)} - \frac{1}{4} \cdot \frac{2x}{(x^2+1)}$$

$$f'(x) = \frac{3}{x+1} - \frac{2x}{4(x^2+1)}$$

$\Rightarrow$ TCD: $4(x+1)(x^2+1)$

$$f'(x) = \frac{12(x^2+1) + (-1)[(2x)(x+1)]}{4(x+1)(x^2+1)}$$

$$\boxed{f'(x) = \frac{5x^2 - x + 6}{2(x+1)(x^2+1)}}$$

Day 15